

# LUT-Fuse Inference Benchmark

CUDA   Batch size 1   4 images   720 x 1280

## 1. Setup

| Item                   | Value                   |
|------------------------|-------------------------|
| Device                 | cuda                    |
| Batch size             | 1                       |
| Measured images        | 4                       |
| Input image size (HxW) | 720 x 1280 (4 image(s)) |

## 2. Latency and Throughput

| Item                       | Value                                   |
|----------------------------|-----------------------------------------|
| Scope                      | apply_fusion_4d_with_interpolation only |
| Excluded                   | data loading, resizing, image saving    |
| Warm-up                    | 1 pass (excluded from measurements)     |
| Total inference time       | 0.040060 sec                            |
| Average latency            | 0.010015 sec/image                      |
| Average latency            | 10.015 ms/image                         |
| Latency standard deviation | 1.035 ms                                |
| Throughput                 | 99.849 FPS                              |

## 3. Parameters

| Item                           | Value                 |
|--------------------------------|-----------------------|
| Context generator parameters   | 7,825 (0.008 M)       |
| Context generator trainable    | 7,825 (0.008 M)       |
| LUT shape                      | 16 x 16 x 16 x 16 x 1 |
| LUT entries                    | 65,536 (0.066 M)      |
| Total inference learned values | 73,361 (0.073 M)      |

## 4. CUDA VRAM

| Item                  | Value                                               |
|-----------------------|-----------------------------------------------------|
| Peak memory allocated | 246.42 MiB                                          |
| Peak memory reserved  | 360.00 MiB                                          |
| Note                  | Max across images, including ckpt and input tensors |

Source: results/benchmark\_metrics.txt